

Professional Summary

AI/ML Engineer specializing in LLM-integrated applications and NLP pipelines, building deployment-ready systems for semantic search, Generative AI APIs, and intelligent automation workflows. Experienced in developing scalable AI & ML solutions with expertise in Python, LLM evaluation, low-latency retrieval systems, and production-ready API deployment.

Skills

Programming Languages: C, C++, Python, Java, SQL, JavaScript

Web Technologies: Django, HTML, CSS, JavaScript, React

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, LLMs, Generative AI APIs (OpenAI, Anthropic), Vector Databases (FAISS), Model Deployment (Flask/FastAPI), Hugging Face, Automation Workflows, Content Personalization

Database: MySQL

Tools & Frameworks: Flask, FastAPI, Pytest, Git, GitHub, Docker, Webhooks, Rest APIs, VS Code, Jupyter Notebook

Projects & Experience

Scalable Semantic Search API (FAISS + FastAPI) - [\[Github Link\]](#)

May 2025 - June 2025

Individual Project - Tech Stack: Python, FastAPI, FAISS, SentenceTransformers, Pytest, Docker, Docker Compose, Uvicorn

- Engineered a **containerized semantic search API** with FAISS vector indexing, delivering **top-K results in <200 ms** across **1K+ documents** with single-load model initialization.
- Achieved **100% test pass rate** through strict request validation and centralized error handling, ensuring deterministic responses and **zero invalid query execution**.
- Delivered **one-command Docker deployment** with environment-driven configuration, enabling **100% reproducible setup** across local and container environments.
- Designed **webhook-ready REST API endpoints** with structured JSON responses, enabling seamless integration with downstream automation pipelines and marketing systems.

Production-Grade LLM Evaluation Framework - [\[Github Link\]](#)

Dec 2025 - Jan 2026

Individual Project - Tech Stack: Python, Poetry, Typer, Pydantic, Pandas, NumPy, NLTK, Transformers, SentenceTransformers, PyTorch (CPU), OpenAI API, Anthropic API, Matplotlib, Seaborn, Pytest, GitHub Actions, Docker, Docker Compose

- Architected a **modular LLM evaluation CLI** with plugin-based metrics applicable to **content quality scoring, marketing copy evaluation, and audience targeting workflows** — enabling **reproducible multi-metric benchmarking** and reducing manual review effort by **100%**.
- Automated **end-to-end evaluation workflows** using Docker Compose and validated YAML configs, achieving **one-command execution** and **<10-minute environment setup** with identical local and container outputs.
- Implemented **retry logic, timeout handling, and graceful degradation** for API and embedding metrics, ensuring **100% pipeline completion** under partial failures with CI-enforced **≥80% test coverage**.
- Deployed **GitHub Actions CI** with automated tests, quality gates, and artifact uploads on every push, delivering **reproducible benchmarking reports** and maintaining a **100% pass-rate validation pipeline**.

Internship Experience

Smartbridge Educational Services Pvt Ltd - [Internship Certificate Received](#)

May 2025 - Jul 2025

- Trained and evaluated **multiple deep learning models** for image classification using **TensorFlow, Keras, and scikit-learn**, applying **CNNs and transfer learning (VGG16, ResNet)** to achieve **85%+ validation accuracy**.
- Implemented **end-to-end ML pipelines** including data preprocessing, model architecture design, hyperparameter tuning, and performance evaluation — translating outputs into **business-relevant metrics**.

Education

B.Tech in Computer Science & Engineering (AIML)

2023 - 2027

Aditya College Of Engineering & Technology, Surampalem

CGPA: 8.33

Certifications

AWS AI Practitioner - Cloud AI & ML Deployment [\[✓\]](#)

Microsoft Azure AI - Enterprise ML solutions and cognitive services [\[✓\]](#)

Python - Advanced python programming and best practices [\[✓\]](#)